Samuel James Baker IV

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EDUCATION

University of Virginia, School of Engineering and Applied Science • 3.5/4.0 **Expected May 2027**
*Bachelor of Science in Computer Science - **Artificial Intelligence Focal Path**, Engineering Business Minor*
Charlottesville, VA

Rappahannock Community College • 4.0/4.0 **Aug 2019 - May 2023**
Associate Arts and Science, Summa Cum Laude
Warsaw, VA

TECHNICAL SKILLS

Languages: C++, C, Python, SQL, TypeScript/JavaScript, Java, C#, Rust
ML/AI: PyTorch, TensorFlow, OpenCV, Hugging Face, Transformers, time-series forecasting, NLP, feature engineering, model training/evaluation, inference optimization, LLM tooling
Stack: React, Next.js, Tailwind CSS, Node.js, Express, REST APIs, MongoDB
Systems/Graphics: OpenGL 4.6, GPU compute, CUDA, Vulkan 1.3, DirectX 11, SDL3, ImGui, Assimp, stb, GLAD, Linux, Git

PROJECTS

Entity Component System Game Engine • Charlottesville, VA **Jun 2025 - Present**
- Developed a **C++20** ECS game engine using **SDL3** and **OpenGL** with an **ImGui** editor, asset pipeline and profiling tools.
- Implemented a **real-time GPU path tracer** via OpenGL compute shaders (**progressive accumulation** and **A-Trous denoising**), achieving **200-40 FPS at 1-8 samples per pixel** on an RTX 4070 laptop GPU.
- Redesigned component storage from hash maps to dense arrays, reducing 100k entity add/remove operations from **99.8ms to 11.8ms (~88% reduction, 8.5x faster)**; and system-update time from **39.8ms to 6.93ms (~83% reduction, 5.7x faster)**; scaled to **multi-million polygon** scenes with **~27.8 ns** component lookup.

Self-Hosted AI Infrastructure Lab • Charlottesville, VA **Jun 2025 - Present**
- Built and operate a **self-hosted homelab** for **local LLM inference** and multiplayer game services, prioritizing **privacy**, transparency, and control over third-party **cloud platforms**.
- Implemented **secure remote access over SSH** and designed architecture for **decentralized self-hosted services**, with expansion paths for **mesh VPN** and **DNS-layer filtering**.
- Developed a **custom UI** for **tokenization** and model interaction to accelerate local inference, **fine-tuning**, and iterative model testing.

Eggs by the Dozen • University of Virginia **Mar 2024**
- **Collaborated** on a 5-member team to build a **Docker**ized full-stack parasite-screening app with **HTML/CSS/JavaScript**, **Express.js**, and **Python/OpenCV**, automating McMaster-slide analysis for sheep and goat samples.
- Designed a **feature-based CV pipeline** using grayscale blur, thresholding, contour detection, and ellipse/shape-color matching to estimate eggs per gram under iPhone 6 and low-connectivity constraints, avoiding **CNNs due to limited labeled data**.
- Improved threshold selection from a fixed value to **modified Otsu with a translational shift**, enabling correct health-condition **classification** on all test samples while reducing testing cost and turnaround versus the manual McMaster workflow.

Agentic RPG Arbiter • Charlottesville, VA **Aug 2025**
- Built a chat-driven RPG arbiter that converts player dialogue into persistent **state flags** and **event triggers**, enabling **branching narrative and adaptive encounters**.
- Added lightweight **memory and session persistence** to maintain narrative continuity across multi-turn interactions in a text-adventure game loop.
- Prototyped **agentic interaction** patterns for games, focusing on stateful control flow and **dialogue-to-action mapping**.

Mycorrhizal Fungi Inoculation in Glycine max. • Chesapeake Bay Governor's School Symposium **Feb 2023**
- Ran controlled inoculation study on soybeans (**n=60**, **6 groups**, **2 trials**) under **70% / 100% / 130%** water-holding capacity over **7-8 weeks**.
- Found significant differences in **shoot dry mass** (**p=0.007-0.023**) and **chlorophyll** (**p=0.014-0.032**); presented at Chesapeake Bay Governor's School symposium (**VCU**).